
Quantum Thermometry in Strongly Coupled and Finite Systems

1 Introduction

When we want to measure the temperature of an everyday object, the process seems simple. We place a thermometer in contact with the object, wait for them to reach thermal equilibrium, and read the number. However, when we scale this process down to the microscopic quantum level, the rules of measurement become much more complicated. The goal of this report is to define exactly what it means for a quantum thermometer to "work well" when it is forced to interact with non-ideal environments.

To study quantum thermometry, we build a simple theoretical model. We define a target system (S) whose temperature we want to measure. We also define a separate, controllable auxiliary quantum system that acts as our thermometer. In the language of quantum metrology, this auxiliary system is referred to as a "quantum probe" (M) (Mehboudi et al., 2019)¹. The term probe is used because this secondary system is explicitly designed to interact with, or "probe", the target system to encode information about the target's temperature into its own quantum state.

In an ideal situation, the system S has an initial, undisturbed temperature. To find out what this temperature is, we must bring the quantum probe M into contact with the system S and let them exchange energy until they settle into a final, steady equilibrium state. By measuring the final state of the probe, we attempt to deduce the original temperature of the system.

The core challenge in quantum thermometry is that measurements are never passive. In classical physics, a thermometer is usually so large that placing it in a cup of coffee does not noticeably change the coffee's temperature. In quantum mechanics, however, to connect the thermometer to the system, we must turn on an interaction Hamiltonian (V).

Because the objects are so small, this interaction energy is often non-negligible. As highlighted by Pernau-Llobet et al. (2018)², this connection inevitably disturbs both the thermometer and the system. The two parts become physically intertwined. Because of this disturbance, a good quantum thermometer cannot simply absorb and perfectly mirror the system's initial state. The act of measuring the system changes the system itself.

Because the interaction V changes the state of our devices, we face a fundamental question: what number should the thermometer show for us to say that it is accurate?

¹Mohammad Mehboudi, Anna Sanpera and Luis A. Correa (2019). [Thermometry in the quantum regime: recent theoretical progress](#).

²M. Pernau-Llobet, H. Wilming, A. Riera, R. Gallego, and J. Eisert (2018). [Strong Coupling Corrections in Quantum Thermodynamics](#).

We can define a successful quantum measurement through a conceptual framework closely tied to *quantum Bayesian retrodiction* (Bai et al., 2023). Applying this to our problem, a quantum thermometer works well if we can look at its final equilibrium state and use statistical retrodiction to calculate backwards. If the final reading gives us enough information to accurately reconstruct what the system's original undisturbed temperature was prior to the coupling, then the thermometer is successful. In this report, we test this definition of success across two scenarios: an infinite system with strong coupling, and a finite system.

2 The Role of Retrodiction in Quantum Thermodynamics

This chapter reviews recent literature to justify why retrodiction is the correct tool for finding the initial state and how it connects to the physical reverse of thermodynamic processes.

2.1 Justifying Retrodiction to Find the Initial State

In classical statistics, if we have a noisy measurement outcome, we can use probability theory to make a guess about the original signal. In quantum mechanics, however, the process is far more complex because the act of measurement actively scrambles the state of the system.

To overcome this hurdle, modern quantum information theory utilizes quantum Bayesian retrodiction. The necessity of this approach in highly disturbing measurement scenarios was recently formalized by Bai et al. (2023). In their study of "observational entropy", Bai and colleagues demonstrated how quantum measurements blur the underlying true state of a physical system, making it mathematically difficult to infer the original input from the final measurement.

This is exactly the hurdle our thermometer faces. By applying the principles of Bayesian retrodiction, often implemented mathematically through recovery operations such as the Petz map, we can systematically process the final state of our probe. Bai et al (2023) show that while we cannot physically undo the measurement, statistical retrodiction provides a mathematical recipe to infer the unknown past input state from a final noisy outcome. For our purposes, this means we can use the final reading of the probe to calculate backward and estimate the undisturbed initial temperature of the system S .

2.2 Retrodiction and Reversing Thermodynamics

While retrodiction is fundamentally a mathematical tool, it has deep physical implications in quantum thermodynamics. If we are using math to go backward in time to find an initial temperature, we must ask: what does this mean for the physical thermodynamics of the system?

In classical thermodynamics, time flows in one direction, and reversing a process usually requires doing work to undo entropy generation. In the quantum regime, the mathematics of retrodiction can be used to define what a physically reversed thermodynamic process actually

looks like. A critical breakthrough in this area was recently presented by Aw et al. (2024)³

Aw and colleagues proved that the statistical recipe for information retrieval is mathematically equivalent to a physical process called "dilation". In a dilation protocol, a physical process is reversed by attaching an auxiliary environment (a bath), applying a reverse global operation, and then discarding the bath. The profound conclusion of Aw et al. is that calculating backward statistically is identical to mathematically simulating this physical reversal of the thermodynamic disturbance.

Therefore, when we apply operational retrodiction to our quantum thermometer, we are not just doing arbitrary statistics. We are effectively applying a formal thermodynamics reversal. By calculating backward, we are mathematically undoing the heat exchange and disturbance cause by the interaction Hamiltonian (V), allowing us to theoretically isolate the initial thermal state of the target system.

3 Methodology and the Evaluation Protocol

To determine if our quantum thermometer is successful, we must establish a rigorous mathematical protocol to simulate the measurement process and evaluate its accuracy. Our strategy is divided into three distinct phases: the thermometer-system interaction, the assignment of a nonequilibrium temperature, and the retrodictive calculation to find the initial state.

3.1 The Thermometer-System Interaction

The first step of the protocol is to model the physical interaction. We initialize the target system (S) at an unknown initial temperature, and prepare our quantum probe (M) in a known reference state. We then allow the two systems to interact via a non-negligible coupling Hamiltonian (V) and evolve the joint system until it reaches a steady state. Finally, we trace out the system to find the final reduced density matrix of the probe, ρ_M^{final} . Because of the strong coupling (in the infinite bath case) or back-action (in the finite bath case), this final state is highly perturbed and correlated with the system.

3.2 Assigning Temperature to Non-Gibbsian States

In traditional thermometry, the final state of the probe would be a canonical Gibbs state, allowing us to easily extract the temperature. However, due to the measurement disturbances in our models, ρ_M^{final} is a nonequilibrium state.

To overcome this, we utilize the framework established by Alipour et al. (2021)⁴. They demonstrated that a rigorous thermodynamic temperature can be assigned to a general nonequilibrium quantum system. By expanding the state based on a complete set of observables, Alipour et al. show that temperature can be defined as the partial derivative of

³Clive Cenxin Aw, Lin Htoo Zaw, Maria Balanzó-Juandó and Valerio Scarani. [Role of Dilations in Reversing Physical Processes: Tabletop Reversibility and Generalized Thermal Operations.](#)

⁴S. Alipour, F. Benatti, M. Afsary, F. Bakhshinezhad, M. Ramezani, T. Ala-Nissila, and A. T. Rezakhani. [Temperature in Nonequilibrium Quantum Systems.](#)

the von Neumann entropy with respect to the internal energy. We will apply this exact definition to our probe's final state, allowing us to extract a formal temperature reading T_M^{final} , even when standard canonical assumptions break down.

3.3 The Retrodiction Protocol

Extracting T_M^{final} is not the end of the process, as this value reflects the disturbed state of the devices, not the original target system. The final step is to apply operational retrodiction. Using the final state parameters obtained via the Alipour et al. framework, we apply a mathematical recovery operation (such as the Petz map) to reverse the thermodynamics disturbance caused by V .

By mapping the final nonequilibrium state backward, we calculate an estimated initial temperature for the system, $T_S^{\text{estimated}}$. The strategy for solving our problem therefore to compare $T_S^{\text{estimated}}$ to the true initial temperature of the system. The mathematical distance between these two values will serve as the ultimate metric for how well the quantum thermometer performs in non-ideal environments.

4 Case One: The Infinite System and Strong Coupling

To test our protocol, we first establish a baseline scenario where the target system S is infinitely large. In traditional thermodynamics, an infinite bath represents an ideal environment: it has an infinite heat capacity, meaning its temperature should remain perfectly constant regardless of what is placed inside it. However, in the quantum regime, we must account for the physical interaction energy between the probe and the bath.

Following the exact framework utilized by Perarnau-Llobet et al.⁵, we model this scenario using the Caldeira-Leggett model. To maintain mathematical exactness and avoid perturbative approximations, we model our quantum probe (S) not as a discrete qubit, but as a central quantum harmonic oscillator. The target system (S) is modeled as a continuous bosonic bath composed of N peripheral harmonic oscillators.

The total Hamiltonian of the combined setup is given by

$$H_{\text{tot}} = H_S + H_B + gV + H_L,$$

where H_L is a renormalization term. Here, the free Hamiltonian of the qubit probe is

$$H_S = \frac{1}{2} (m\omega_0^2 x^2 + p^2/m), \quad (1)$$

where x and p are the canonical position and momentum operators. The infinite bath is described by

$$H_B = \frac{1}{2} \sum_{\mu} \left(m_{\mu} \omega_{\mu}^2 x_{\mu}^2 + p_{\mu}^2 / m_{\mu} \right). \quad (2)$$

⁵M. Perarnau-Llobet, H. Wilming, A. Riera, R. Gallego, and J. Eisert. [Strong Coupling Corrections in Quantum Thermodynamics](#).

The crucial element of this model is the interaction Hamiltonian, V . We again assume the same coupling as the Perarnau-Llobet et al. paper, which is

$$V = x \sum_{\mu} g_{\mu} x_{\mu}. \quad (3)$$

The parameter g dictates the overall strength of this interaction. In the CL model, the algebraic necessity of the renormalization term, H_L , becomes clear when we combine the physical Hamiltonian of the environment and the interaction. If we sum the bath's potential energy, $\sum_k m_k \omega_k^2 x_k^2 / 2$, and the interaction energy, $gx \sum_k g_k x_k$, we can algebraically complete the square for the bath coordinates x_k .

$$E_{\text{pot}} = \frac{1}{2} m_k \omega_k^2 x_k^2 + g g_k x x_k$$

To complete the square, we factor out $m_k \omega_k^2 / 2$:

$$\begin{aligned} E_{\text{pot}} &= \frac{1}{2} m_k \omega_k^2 \left(x_k^2 + \frac{2g g_k}{m_k \omega_k^2} x x_k \right) = \frac{1}{2} m_k \omega_k^2 \left[\left(x_k + \frac{g g_k}{m_k \omega_k^2} x \right)^2 - \left(\frac{g g_k}{m_k \omega_k^2} x \right)^2 \right] \\ &= \frac{1}{2} m_k \omega_k^2 \left(x_k + \frac{g g_k}{m_k \omega_k^2} x \right)^2 - \frac{g^2 g_k^2}{2 m_k \omega_k^2} x^2. \end{aligned} \quad (4)$$

To ensure the thermometer's physical frequency is not distorted by this shift, the counter-term H_L must perfectly cancel it out. Therefore,

$$H_L = x^2 g^2 \sum_{\mu} \frac{g_{\mu}^2}{2 m_{\mu} \omega_{\mu}^2}. \quad (5)$$

We will apply the strategy described before this to see if we can un-entangle the reading to find the bath's true initial temperature.

4.1 The Thermometer-System Interaction phase

The state of the entire "universe" at equilibrium is given by the global density operator:

$$\rho_{\text{tot}}^{\text{final}} = \frac{e^{-\beta H_{\text{tot}}}}{\mathcal{Z}_{\text{tot}}} = \frac{e^{-\beta(H_S + H_B + gV + H_L)}}{\mathcal{Z}_{\text{tot}}}, \quad (6)$$

where $\mathcal{Z}_{\text{tot}} = \text{Tr}_{MS}(e^{-\beta H_{\text{tot}}})$ is the global partition function, ensuring the probabilities sum to one.

The total Hamiltonian is

$$H_{\text{tot}} = \frac{p^2}{2m} + \frac{1}{2} m \omega_0^2 x^2 + \sum_{\mu=1}^N \left[\frac{p_{\mu}^2}{2m_{\mu}} + \frac{1}{2} m_{\mu} \omega_{\mu}^2 \left(x_{\mu} + \frac{g g_{\mu}}{m_{\mu} \omega_{\mu}^2} x \right)^2 \right]. \quad (7)$$

We define a unified vector of coordinates that groups the position and momentum operators of the harmonic oscillator and the n modes:

$$\mathbf{r} = (x, x_1, x_2, \dots, x_n, p, p_1, \dots, p_n)^T. \quad (8)$$

We want to show that the total Hamiltonian is equivalent to the quadratic form $H = \frac{1}{2} \mathbf{r}^T \mathbf{H}_r \mathbf{r}$ by expanding the matrix and recovering the original physical term.

Since the entries of $\mathbf{r}$ are Hermitian observables, the transpose operation is equivalent to the adjoint, allowing us to write

$$H = \frac{1}{2} \mathbf{r}^\dagger \mathbf{H}_r \mathbf{r}.$$

The total Hamiltonian contains solely squared terms and position-position cross terms (xx_μ). Because there are no products coupling a position with a momentum, the matrix $\mathbf{H}_r$ must take a block-diagonal structure,

$$\mathbf{H}_r = \begin{pmatrix} \mathbf{U} & 0 \\ 0 & \mathbf{K} \end{pmatrix}. \quad (9)$$

To find the kinetic matrix $\mathbf{K}$, we isolate all momentum terms from H_S and H_B to get

$$E_{\text{kin}} = \frac{1}{2} \left(\frac{1}{m} p^2 + \sum_{\mu} \frac{1}{m_{\mu}} p_{\mu}^2 \right). \quad (10)$$

Mapping this to the momentum vector $\mathbf{r}_p = (p, p_1, \dots, p_n)^T$ yields a purely diagonal matrix:

$$\mathbf{K} = \text{diag} \left(\frac{1}{m}, \frac{1}{m_1}, \dots, \frac{1}{m_N} \right). \quad (11)$$

Next, we isolate all potential energy terms, including the physical springs, the interaction and the shift:

$$E_{\text{pot}} = \frac{1}{2} m \omega^2 x^2 + \frac{1}{2} \sum_{\mu} m_{\mu} \omega_{\mu}^2 x_{\mu}^2 + g x \sum_{\mu} g_{\mu} x_{\mu} + x^2 g^2 \sum_{\mu} \frac{g_{\mu}^2}{2 m_{\mu} \omega_{\mu}^2}.$$

Factoring out the $\frac{1}{2}$, we get

$$E_{\text{pot}} = \frac{1}{2} \left[\left(m \omega^2 + g^2 \sum_{\mu} \frac{g_{\mu}^2}{m_{\mu} \omega_{\mu}^2} \right) x^2 + \sum_{\mu} (m_{\mu} \omega_{\mu}^2) x_{\mu}^2 + 2 \sum_{\mu} (g g_{\mu}) x x_{\mu} \right]. \quad (12)$$

From this polynomial we can directly read the matrix elements. The top left diagonal element corresponding to x^2 is $m \omega^2 + g^2 \sum_{\mu} \frac{g_{\mu}^2}{m_{\mu} \omega_{\mu}^2}$, the remaining diagonal elements corresponding to x_{μ}^2 are $m_{\mu} \omega_{\mu}^2$, and the off-diagonal cross terms are shared symmetrically, meaning the entry

for xx_μ is gg_μ . Therefore, the matrix U is something like this:

$$U = \begin{pmatrix} m\omega^2 + g^2 \sum_\mu \frac{g_\mu^2}{m_\mu \omega_\mu^2} & gg_1 & gg_2 & \cdots & gg_n \\ gg_1 & m_1 \omega_1^2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ gg_n & 0 & 0 & \cdots & m_n \omega_n^2 \end{pmatrix}. \quad (13)$$

Therefore we can write the total Hamiltonian as

$$H = \frac{1}{2} \mathbf{r}^\dagger \mathbf{H}_r \mathbf{r}. \quad (14)$$

In this form, the vector $\mathbf{r}$ contains all the physical position and momentum operators, while the matrix $\mathbf{H}_r$ contains the numerical coupling constants, masses, and frequencies. The position and momentum variables inside $\mathbf{r}$ are mathematically entangled solely because $\mathbf{H}_r$ contains non-zero off-diagonal elements. At this point, we apply the Williamson theorem to matrix $\mathbf{H}_r$ because modifying the numerical coefficients is the exact mathematical mechanism that turns the original physical variable $\mathbf{r}$ into the decoupled normal-mode variables ($\mathbf{q}$).

But why do we need to use the Williamson theorem? Why not use a standard orthogonal matrix? From a purely mathematical standpoint, standard linear algebra diagonalizes a symmetric matrix using an orthogonal matrix. However, applying an arbitrary orthogonal transformation to a physical system treats all $2n$ dimensions identically. This mathematical mixing destroys the fundamental Poisson brackets $\{q_i, p_k\} = \delta_{jk}$ (or canonical commutators $[\hat{q}_j, \hat{p}_k] = i\hbar \delta_{jk}$). Williamson theorem provides the solution to this problem.

The following subsection explains Williamson's theorem, following the presentation by F. Nicacio in [Williamson theorem in classical, quantum, and statistical physics](#).

4.1.1 Williamson Theorem

To understand this theorem, we first need to define a few concepts. Two matrices $\mathbf{A}, \mathbf{B} \in \mathbf{M}(n)$ are congruent if there is an invertible matrix $\mathbf{C} \in \mathbf{M}(n)$ such that:

$$\mathbf{A} = \mathbf{C} \mathbf{B} \mathbf{C}^\dagger.$$

Here, $\mathbf{M}(n)$ denotes the set of all $n \times n$ complex square matrices, and $\mathbf{M}(n, \mathbb{R})$ denotes the set of real matrices. By Sylvester's law of inertia, the inertia of $\mathbf{A}$ and $\mathbf{B}$ is the same if and only if these matrices are Hermitian. The inertia is defined as follows: Let $A \in M_n$ be a Hermitian matrix. The inertia of A is the ordered triple:

$$i(A) = (i_+(A), i_-(A), i_0(A))$$

where $i_+(A)$, $i_-(A)$, and $i_0(A)$ are the number of positive, negative, and zero eigenvalues of A , respectively⁶.

⁶R. A. Horn and C. R. Johnson, Matrix Analysis, 2nd ed. (Cambridge University Press, 2013), Definition 4.5.6

A symplectic matrix $S \in \mathbf{M}(2n, \mathbb{R})$ is defined by the rule

$$S^T J S, \quad \text{where } J = \begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix} \in \mathbf{M}(2n, \mathbb{R}). \quad (15)$$

The matrix J is such that $J^2 = -I_{2n}$, thus $J^{-1} = -J = J^T$, and itself a symplectic matrix with $\det(J) = 1$. Taking the transposition of S , one can show that S^T is also symplectic and that the condition $SJS^T = J$ is equivalent to Eq. (15). The determinant of a symplectic matrix, from the definition, is such that $\det(S^2) = 1$. Consequently, every symplectic matrix is invertible and the inverse is $S^{-1} = J^T S^T J$ by Eq. (15). Finally, the set of symplectic matrices forms the group

$$S_p(2n, \mathbb{R}) := \{S \in \mathbf{M}(2n, \mathbb{R}) \mid S^T J S = J\}.$$

Now we can finally get to the theorem itself.

Theorem: Let $\mathbf{M} \in \mathbf{M}(2n, \mathbb{R})$ be symmetric and positive-definite, i.e., $\mathbf{M}^T = \mathbf{M} > 0$. There exists $S_M \in S_p(2n, \mathbb{R})$ such that

$$S_M \mathbf{M} S_M^T = \Lambda_M \quad \Lambda_M := \text{Diag}(\mu_1, \dots, \mu_n, \mu_1, \dots, \mu_n), \quad (16)$$

with $0 < \mu_j \leq \mu_k$ for $j \leq k$. Each μ_j is such that

$$\det(\mathbf{J}\mathbf{M} \pm i\mu_j \mathbf{I}_{2n}) = 0 \quad (j = 1, \dots, n), \quad (17)$$

and the matrix S_M admits the decomposition

$$S_M = \sqrt{\Lambda_M} O \sqrt{\mathbf{M}^{-1}}, \quad (18)$$

where $O \in \mathbf{M}(2n, \mathbb{R})$ satisfies

$$O \sqrt{\mathbf{M}} J \sqrt{\mathbf{M}} O^T = \Lambda_M J, \quad (19)$$

and $O^T = O^{-1}$, i.e., is an orthogonal matrix.

The proof of this theorem is beyond the scope of this report.

Invoking Williamson theorem on $\mathbf{H}_r$, we discover a specific symplectic matrix S that decomposes the coefficient matrix into a diagonal form:

$$\mathbf{H}_r = S^\dagger (D \oplus D) S. \quad (20)$$

Substituting this back into the total Hamiltonian equation gives

$$\mathbf{H}_{\text{tot}} = \frac{1}{2} \mathbf{r}^\dagger S^\dagger (D \oplus D) S \mathbf{r}. \quad (21)$$

Because matrix multiplication is associative, we can group the symplectic matrix S with our original operator $\mathbf{r}$. We define a new vector of transformed coordinates as $\mathbf{q} = S \mathbf{r}$. Substi-

tuting $\mathbf{q}$ into the equation yields

$$\mathbf{H}_{\text{tot}} = \frac{1}{2} \mathbf{q}^\dagger (D \oplus D) \mathbf{q}. \quad (22)$$

To the current coordinates to ladder operators, we introduce the transformation matrix

$$\Omega = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ -iI & iI \end{pmatrix}. \quad (23)$$

This specific matrix Ω dictates the relationship between position/momentum and the ladder operators. The top block matrix (I) corresponds to the physical definition where position is proportional to the sum of the creation and annihilation operators, while the bottom block matrix ($\pm iI$) corresponds to momentum being proportional to their complex difference.

Using this matrix, we define a new set of independent bosonic operators as $\mathbf{b} = \Omega^{-1} \mathbf{S} \mathbf{r}$. Applying this transformation directly gives rise to the decoupled, exact Hamiltonian

$$\mathbf{H}_{\text{tot}} = \sum_k d_k \left(b_k b_k^\dagger + \frac{1}{2} \right). \quad (24)$$

To find the final state of the thermometer alone, we must isolate it from the environment. Mathematically, we do this by taking the partial trace over the degrees of freedom of the infinite bath $\mathbf{B}$. This yields the reduced density matrix of the probe:

$$\rho_S^{\text{final}} = \text{Tr}_B \left(\rho_{\text{tot}}^{\text{final}} \right) = \frac{1}{\mathcal{Z}_{\text{tot}}} \text{Tr}_B \left(e^{-\beta(H_S + H_B + V)} \right). \quad (25)$$

Because the thermometer Hamiltonian H_S (which contains σ_z) does not commute with the interaction Hamiltonian V (which contains σ_x), we cannot simply factorize the exponential into $e^{-\beta H_M} e^{-\beta H_S} e^{-\beta V}$. This exponential must be expanded perturbatively in powers of g up to $\mathcal{O}(g^2)$. Utilizing first-order Dyson series in imaginary time, the unnormalized thermal state expands as

$$e^{-\beta_0(H_0 + gV)} = e^{-\beta_0 H_0} \left(I - \beta_0 g \int_0^1 d\tau e^{\beta_0 \tau H_0} V e^{-\tau \beta_0 H_0} \right) + \mathcal{O}(g^2), \quad (26)$$

where $\beta_0 = 1/T_0$ is the initial universe temperature and $H_0 = H_S + H_B$. To simplify, we define the integral over the imaginary time variable τ as the operator

$$V_{H_0, \beta_0} = \int_0^1 d\tau e^{\beta_0 \tau H_0} V e^{-\tau \beta_0 H_0}. \quad (27)$$

Substituting this notation, the unnormalized global state simplifies to

$$e^{-\beta_0 H_0} (I - \beta_0 g V_{H_0, \beta_0}) + \mathcal{O}(g^2).$$

The total partition function is obtained by taking the trace of the unnormalized state

over the joint Hilbert space,

$$\mathcal{Z}_{\text{tot}} = \text{Tr} \left(e^{-\beta_0 H_0} \right) - \beta_0 g \text{Tr} \left(e^{-\beta_0 H_0} V_{H_0, \beta_0} \right) + \mathcal{O}(g^2). \quad (28)$$

Defining the uncoupled partition function as

$$\mathcal{Z}_0 = \text{Tr}(e^{-\beta_0 H_0}),$$

the thermal state of the unperturbed system is

$$\omega_{\beta_0}(H_0) = \frac{1}{\mathcal{Z}_0} e^{-\beta_0 H_0}.$$

Factoring out $\mathcal{Z}_0$ yields

$$\mathcal{Z}_{\text{tot}} = \mathcal{Z}_0 \left(1 - \beta_0 g \text{Tr} \left(\omega_{\beta_0}(H_0) V_{H_0, \beta_0} \right) + \mathcal{O}(g^2) \right). \quad (29)$$